

## Multiple Choice (10 marks)

1. Which of the following can only be used when training data are linearly separable?
  - (a) Linear hard-margin SVM.
  - (b) Linear Logistic Regression.
  - (c) Linear Soft margin SVM.
  - (d) The centroid method.
2. Statement 1| L 2 regularization of linear models tends to make models more sparse than L 1 regularization. Statement 2| Residual connections can be found in ResNets and Transformers.
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True
3. What is the dimensionality of the null space of the following matrix?  $A = \begin{bmatrix} 1, & 1, & 1 \\ 1, & 1, & 1 \\ 1, & 1, & 1 \end{bmatrix}$ 
  - (a) 0
  - (b) 1
  - (c) 2
  - (d) 3
4. MLE estimates are often undesirable because
  - (a) they are biased
  - (b) they have high variance
  - (c) they are not consistent estimators
  - (d) None of the above
5. Which image data augmentation is most common for natural images?
  - (a) random crop and horizontal flip
  - (b) random crop and vertical flip
  - (c) posterization
  - (d) dithering

## True / False (10 marks)

*Answer True or False*

6. True or False: Adding more basis functions in a linear model decreases variance by simplifying the model's structure.
7. True or False: ImageNet contains images of various resolutions, making it a diverse dataset for training deep learning models.
8. True or False: Discriminative approaches model the joint probability  $p(y, x)$ , which captures both the output and input distributions.
9. True or False: Principal Component Analysis (PCA) is an unsupervised learning technique that reduces dimensionality without labeled output.
10. True or False: Overfitting is more likely when the set of training data is small, as the model may learn noise instead of the underlying pattern.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to find the  $n$ 'th perrin number using recursion.
12. Write a python function to get the first element of each sublist.

*End of Paper*